
CAFE SIMULATION REPORT

Operational Optimization of a Cafe System using Arena Simulation

Course: CENG 4513 – Modeling and Simulation

Department: Computer Engineering

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Instructor: İlke Tan

1. Introduction & Process Explanation

1.1. Project Objective

The main objective of this project is to model the operational processes of a café by using the ARENA simulation software, analyze the performance metrics of the existing system, and propose improvement strategies in order to increase overall resource utilization efficiency.

The sub-objectives of the project are outlined below:

Process Modelling:

To analyze the real-life workflow of the café in detail, including customer arrivals, order taking, preparation, and service processes, and to represent this workflow accurately within a simulation environment.

Data-Driven Analysis:

To determine appropriate statistical distributions based on real system data and to construct a mathematical representation of the café operations using these distributions.

Performance Evaluation

To evaluate system performance by analyzing key operational metrics, including:

- Measurement of customer waiting times
- Analysis of cashier and barista utilization rates
- Assessment of system efficiency using quantitative performance indicators

Improvement Scenarios

To develop and test alternative scenarios aimed at improving system performance, such as:

- Optimization of resources (staff allocation)
- Redesign of operational processes
- Queue management and prioritization strategies

Comparative Analysis

To compare the performance results of the initial simulation model with those of the improved model and evaluate the impact of the proposed improvements on the overall system.

1.2. Overview of Project Report Content

1.3 Overview of Project Report Structure

The content of this report is organized into ten main sections as follows:

Introduction & Process Explanation: Defines the project objectives, scope, and the overall roadmap of the study.

Data Collection: Details the field observation methodology at Mono Cafe and the types of data collected.

Input Data Analysis: Presents the Arena Input Analyzer results, including statistical distribution selection and rationale.

Initial Model: Explains the technical development and performance results of the baseline simulation.

Improved Model: Details the optimization strategies, multitasking logic, and the results of the enhanced system.

Conclusion and Final Recommendation: Summarizes the comparative findings and provides strategic business advice.

Appendices: Contains the raw observation logs, processed datasets, and detailed Arena output summaries.

2. Data Collection

2.1. Data Collection Methodology

Why this cafe?

Mono Cafe was selected for data collection because:

- Single cashier and single barista configuration (matches our model assumptions)
- Observable workflow (open counter design) -
- Moderate customer traffic (allows accurate timing) -
- Management permission for observation.

How Data Was Collected:

1. **Observation Period:** The data was collected on 14.12.2025, during a continuous session from 11:25:22 to 21:15:30 (totaling 590 minutes).
2. **Collection Method:** A single observer (the researcher) conducted the study to ensure consistency in data classification and recording across all system stages.
3. **Recording Tools:** - A digital stopwatch with millisecond precision was used for timing interactions.
 - Real-time manual logging was performed, then transferred to Microsoft Excel for statistical preprocessing.
 - An observation checklist was utilized to categorize customers based on order types (Drink(1), Food(3), or Both(2)).
4. **Tracking Process:** - Each customer was tracked individually from the moment of entry (Arrival) to the final receipt of their order (System Exit).
 - Interarrival times, service times at the cashier, and specific preparation times by the barista were recorded for 80 distinct customer groups.

5. Traffic Conditions:

- **Peak Intervals:** Significant customer density was observed during lunch hours (12:00-13:30) and evening hours (18:30-20:00), accounting for the busiest periods of the simulation.
- **Operational Flow:** Despite varying traffic intensity, the observation was conducted without interruption to capture the stochastic nature of customer arrivals, later modeled with an Exponential distribution (EXPO 7.38).
- **Environmental Factors:** The study was conducted under normal operating conditions with no external disruptions or equipment failures, ensuring the data represents a standard business day.

Data Types

The data collected and their definitions, which are used to explain the types of data employed in the simulation model and to accurately represent the operation of the system, are presented below.

Arrival Time:

The time at which a customer or customer group arrives at the café system.

Group Size:

The number of customers arriving together as a group.

Order Type:

The category of the order placed by the customer (drinks only, food only, or food and drinks)

Order Start Time:

The time at which the order-taking process begins at the cashier.

Order End Time:

The time at which the order-taking process is completed.

Preparation Start Time:

The time at which the preparation of the order starts.

Preparation End Time:

The time at which the preparation of the order is completed.

Customer Exit Time:

The time at which the customer leaves the system after receiving the order.

Table 1: Field Observation Data Details

Data Type	Collected Data	Data Collection Method	Collection Period	Total Observation
Arrival Time	Customer Arrival Time	Mobile device time tracking (hh:mm:ss) & Excel logging	14.12.2025	80
Group Size	Number of customers per group	Manual counting & Excel logging	14.12.2025	80
Order Type	Order category (drinks only, food only, or food and drinks)	Manual recording & Excel logging	14.12.2025	80
Order Start Time	Start time of order-taking process	Mobile device time tracking (hh:mm:ss) & Excel logging	14.12.2025	80
Order End Time	End time of order-taking process	Mobile device time tracking (hh:mm:ss) & Excel logging	14.12.2025	80
Preparation Start Time	Start time of order preparation	Mobile device time tracking (hh:mm:ss) & Excel logging	14.12.2025	80

Preparation End Time	End time of order preparation	Mobile device time tracking (hh:mm:ss) & Excel logging	14.12.2025	80
Customer Exit Time	Time customer leaves the system	Mobile device time tracking (hh:mm:ss) & Excel logging	14.12.2025	80

2.1. Raw Data Summary

CustomerID	Arrival Time	Group Size	Order Type	Order Start Time	Order End Time	Preparation Start Time	Preparation End Time	System Exit Time
1	11:25:22	1	1	11:27:02	11:27:21	11:27:21	11:27:47	11:28:00
2	11:26:11	1	1	11:31:44	11:32:31	11:32:33	11:35:31	11:35:50
3	11:43:15	2	1	11:43:30	11:44:17	11:44:18	11:48:02	11:48:22
4	12:35:15	1	1	12:35:44	12:36:30	12:36:30	12:38:50	12:40:36
5	12:35:15	1	1	12:35:44	12:36:50	12:36:50	12:40:14	12:40:36
6	12:35:15	1	1	12:35:44	12:37:24	12:37:24	12:41:10	12:40:36
7	12:35:15	1	1	12:35:44	12:37:54	12:37:54	12:42:19	12:40:36
8	12:59:29	1	1	12:59:34	12:59:42	12:59:45	13:01:49	13:02:01
9	13:04:35	1	1	13:04:48	13:05:44	13:05:45	13:06:36	13:06:51
10	13:07:52	1	1	13:07:59	13:08:30	13:08:30	13:09:25	13:09:35

Figure 1. Sample of Collected Data

Note: The table below presents the first 10 rows of the collected dataset. The complete dataset is provided in the Appendix.

2.2. Data Processing and Transformation

The raw time data were collected in hh:mm:ss format during real-time observation. To ensure compatibility with the ARENA simulation environment, all time values were converted into minutes, while the original raw data were preserved and provided in the Appendix.

Interarrival times (IAT) were calculated as the time differences between consecutive customer arrivals. Cashier service times were obtained by subtracting order start times from order end times. Preparation times were calculated as the difference between preparation start and end times and were analyzed separately based on order type (drinks only, food only, and food & drinks).

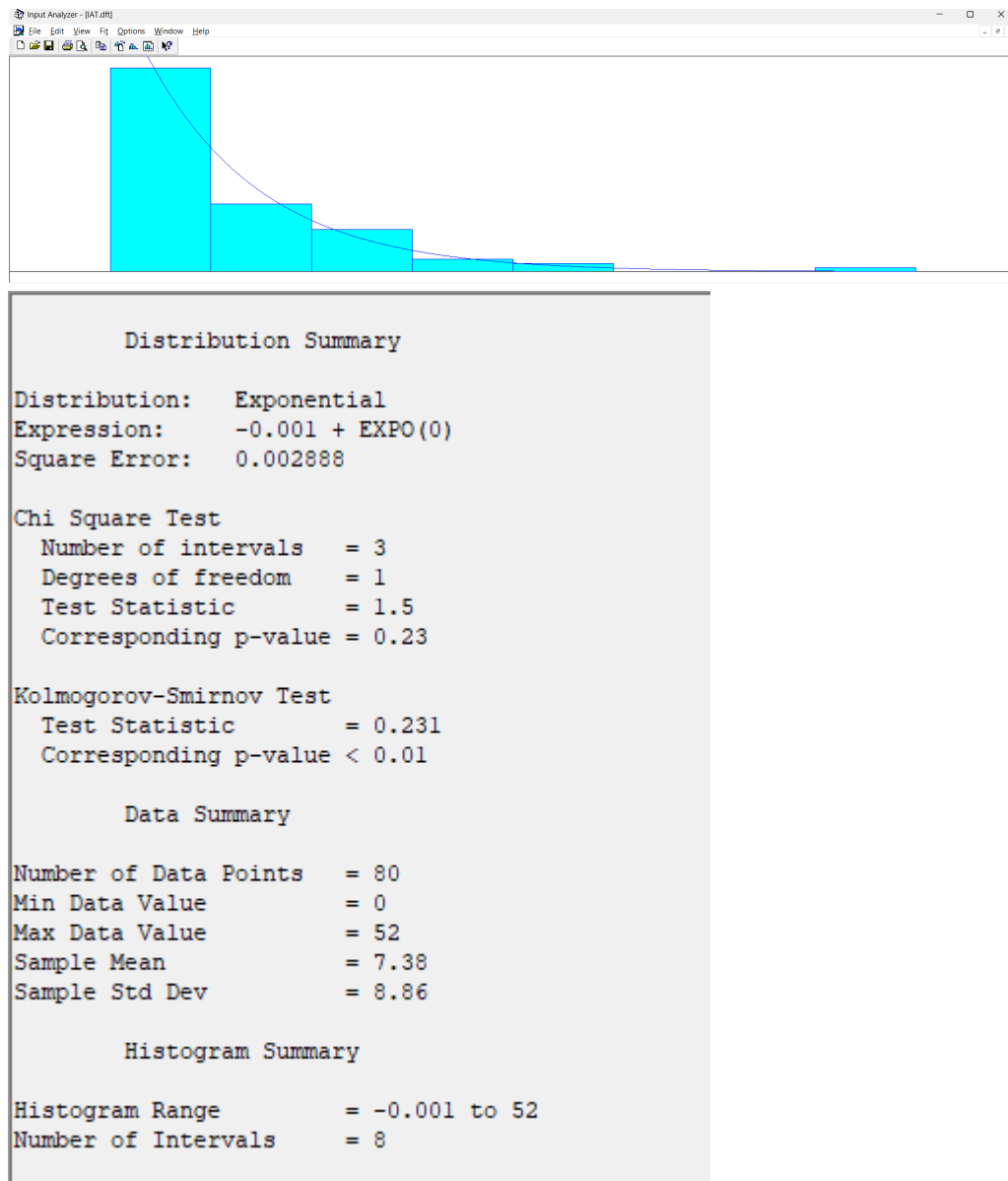
This classification allowed different preparation processes to be represented with distinct statistical distributions. All derived datasets were analyzed using Arena Input Analyzer, and the resulting distributions were implemented in the simulation model.

3. Input Data Analysis

3.1. Statistical Distribution Analysis

The Input Analyzer histograms, fitted distribution graphs, and goodness-of-fit tables are included in this section to justify the selected statistical distributions used in the simulation.

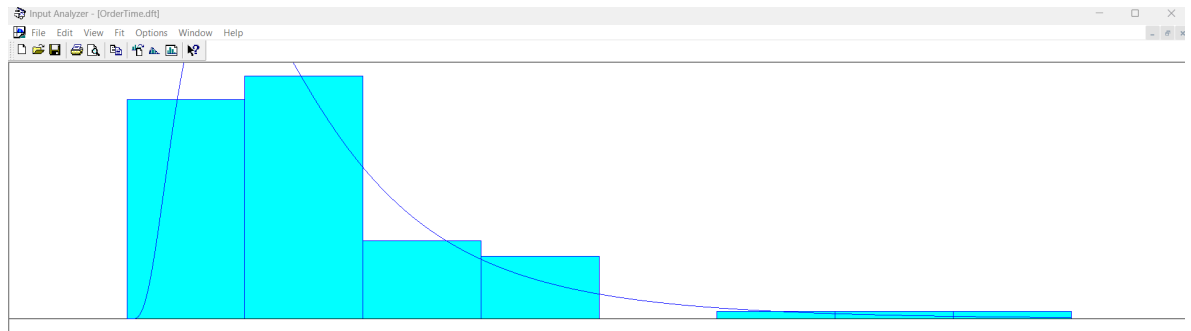
3.1.1. Customer Interarrival Times



Arena Input Analyzer Results:

- **Selected Distribution:** Exponential Distribution
- **Expression:** $-0.001 + \text{EXPO}(7.38)$
- **Square Error:** 0.002888 (low error margin, indicating a good fit)
- **Chi-Square Test:** p-value = 0.23 (> 0.05 , therefore the null hypothesis cannot be rejected)

3.1.2. Customer Order Time



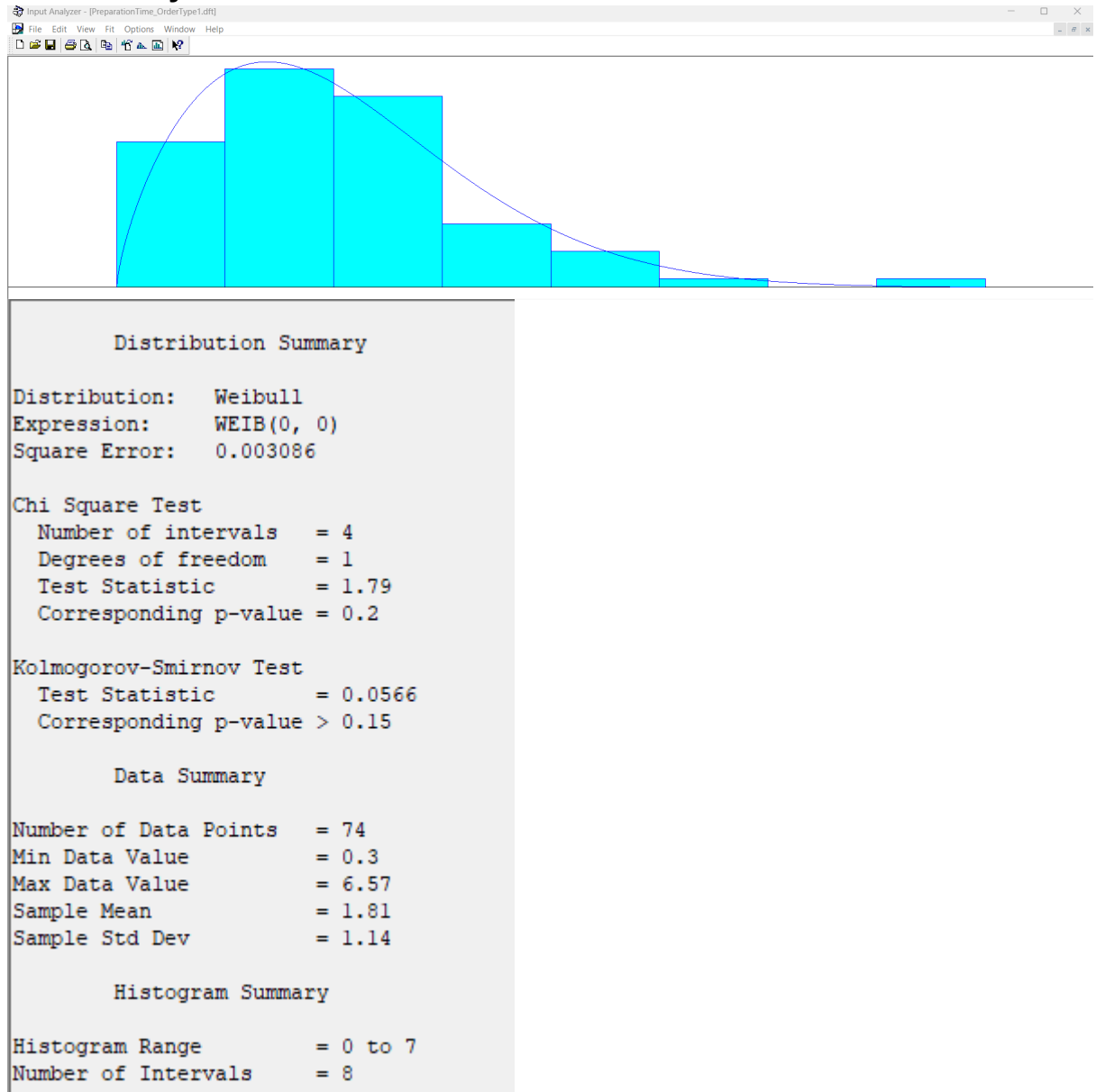
Distribution Summary	
Distribution:	Lognormal
Expression:	LOGN(0, 0)
Square Error:	0.003328
Chi Square Test	
Number of intervals	= 4
Degrees of freedom	= 1
Test Statistic	= 1.03
Corresponding p-value	= 0.334
Kolmogorov-Smirnov Test	
Test Statistic	= 0.0576
Corresponding p-value	> 0.15
Data Summary	
Number of Data Points	= 80
Min Data Value	= 0.1
Max Data Value	= 2.17
Sample Mean	= 0.504
Sample Std Dev	= 0.402
Histogram Summary	
Histogram Range	= 0 to 2.38
Number of Intervals	= 8

Arena Input Analyzer Results:

- **Selected Distribution:** Lognormal Distribution
- **Expression:** LOGN(0.499, 0.389)
- **Square Error:** 0.003328 (Low goodness-of-fit error)
- **Chi-Square Test:** p-value = 0.334
- **Kolmogorov-Smirnov Test:** > 0.15

3.1.3. Customer Preparation Times

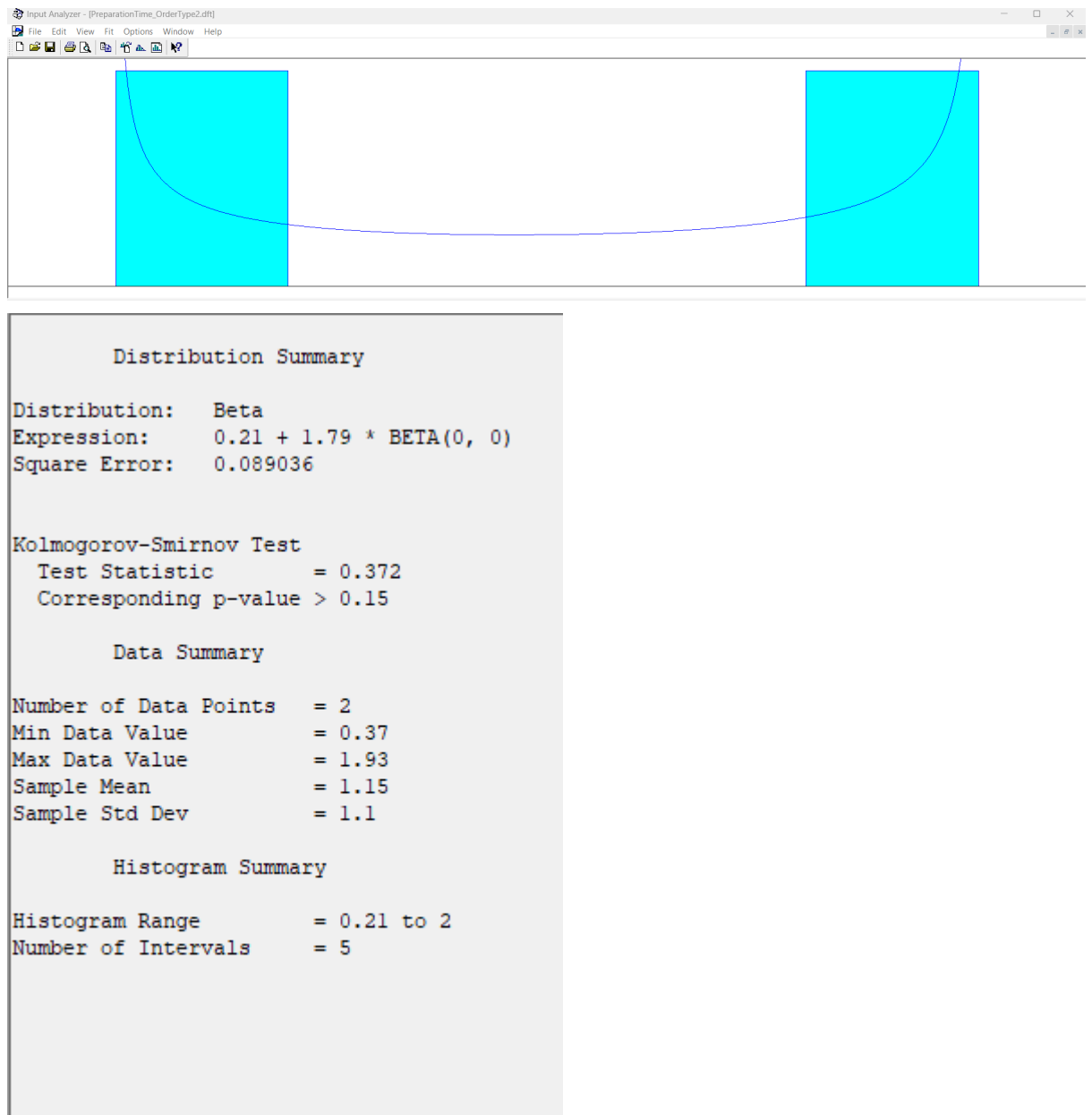
Drinks Only



Arena Input Analyzer Results:

- **Selected Distribution:** Weibull Distribution
- **Arena Expression:** WEIB(2.04, 1.7)
- **Square Error:** 0.004215
- **Chi-Square Test:** p-value > 0.15
- **Kolmogorov-Smirnov Test:** > 0.15

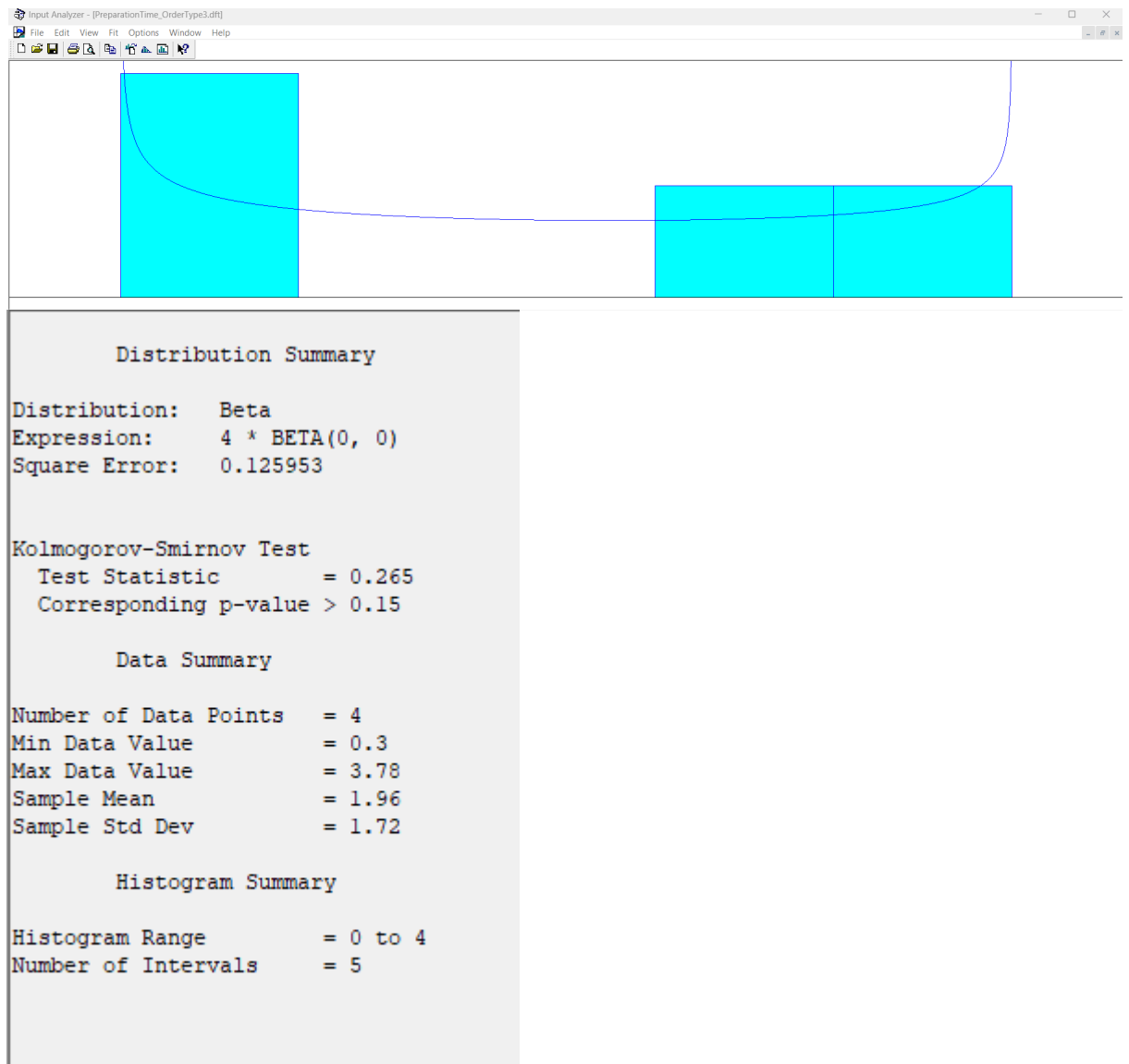
Foods Only



Arena Input Analyzer Results:

- **Selected Distribution:** Beta Distribution
- **Arena Expression:** $0.21 + 1.79 * \text{BETA}(0.00001, 0.00001)$

Food and Drinks



Arena Input Analyzer Results:

- **Selected Distribution:** Beta Distribution
- **Arena Expression:** $4 * \text{BETA}(0.00001, 0.00001)$

Table 2: Preparation Time Analysis Results

Order Type	Distribution	Arena Expression	Sample Size	Cumulative Rate
1) Drinks Only	Weibull	WEIB(2.04, 1.7)	74	0.925
2) Food & Drinks	Beta	$0.21 + 1.79 * \text{BETA}(0.00001, 0.00001)$	2	0.950
3) Food Only	Beta	$4 * \text{BETA}(0.00001, 0.00001)$	4	1.000
Total			80	

3.2. Distribution Selection and Rationale

The selection of probability distributions was performed using Arena Input Analyzer, based on goodness-of-fit measures such as Square Error and p-values from statistical tests.

Interarrival Time – Exponential (-0.001 + EXPO(7.38))

Customer arrivals in café systems can be modeled as independent events occurring at a constant average rate. The Exponential distribution is the standard choice for representing interarrival times in service systems and provided a statistically acceptable fit.

Order Time (Cashier) – Lognormal (LOGN(0.499, 0.389))

Cashier service times are strictly non-negative and exhibit right-skewed behavior, where most transactions are completed quickly while some take longer. The Lognormal distribution accurately represents this variability and achieved a good fit with the observed data.

Preparation Times – Weibull and Beta

Drink preparation times were best represented by the Weibull distribution (WEIB(2.04, 1.7)), due to its flexibility in modeling technical processes with varying durations.

Food-only and combined food-and-drink orders were modeled using Beta distributions, as these processes have well-defined minimum and maximum preparation limits. (The Beta distribution parameters for food preparation were adjusted to 0.00001 instead of 0 to prevent computational errors in Arena while maintaining the statistical integrity of the observed data.)

Group Size and Order Type – Discrete (DISC)

Since group size and order type are categorical variables, Discrete distributions based on empirical probabilities from the observed data were used.

4. Initial Model

4.1. Model Description

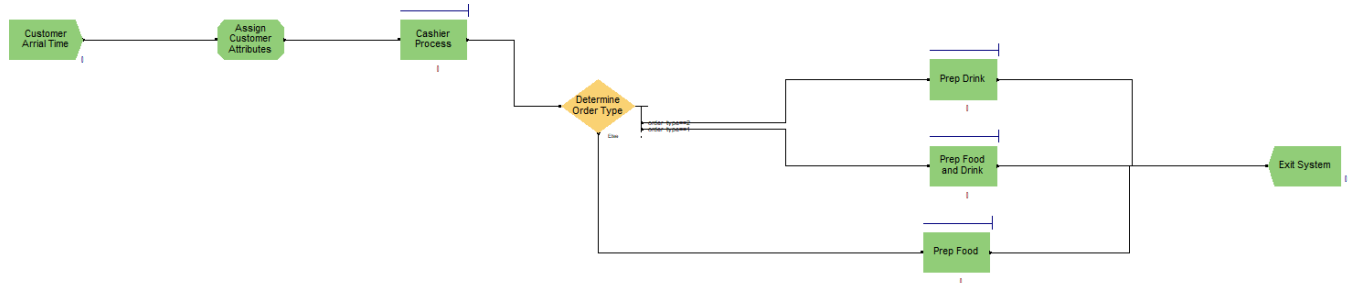


Figure 2. Initial Arena model

The cafe operates as follows:

- Customers arrive according to exponential distribution
- They go to cashier to place order
- Cashier takes the order (Lognormal time)
- Based on order type, customers are directed to preparation:
 - Drink only → Barista (Type 1)
 - Food + Drink → Barista (Type 2)
 - Food only → Barista (Type 3)
- Customer exits after receiving order

Table 3: Detailed Overview of the Simulation Model Components

Module Name	Module Type	Purpose	Key Parameters & Expressions
Customer Arrival Time	Create	Simulates customer entry into the cafe	Interarrival Time: $-0.001 + \text{EXPO}(7.38)$ Minutes

Assign Customer Attributes	Assign	Assigns order type and group size	order_type: DISC(0.925, 1, 0.95, 2, 1.0, 3) group_size: DISC(0.875, 1, 0.975, 2, 0.9875, 4, 1.0, 5)
Cashier Process	Process	Order taking and payment at the counter	Expression: LOGN(0.499, 0.389) Minutes Resource: Cashier (Capacity: 1)
Determine Order Type	Decide	Routes entities based on order_type	Type: N-way by Condition Conditions: order_type == 1 and order_type == 2
Prep Drink	Process	Preparation stage for drink orders	Expression: WEIB(2.04, 1.7) Minutes Resource: Barista (Capacity: 1)
Prep Food and Drink	Process	Preparation for combined orders	Expression: $0.21 + 1.79 * \text{BETA}(0.00001, 0.00001)$ Resource: Barista (Capacity: 1)

Prep Food	Process	Preparation stage for food orders	Expression: 4 * BETA(0.00001, 0.00001) Resource: Barista (Capacity: 1)
Exit System	Dispose	Entities leaving the system	Statistics: Record Entity Statistics enabled

4.2. Run Settings

To obtain statistically reliable results and reflect the real-world operational timeframe of the cafe, the following settings were configured in the **Arena Run Setup** menu:

- **Number of Replications:** The simulation was run for **10 replications**. This ensures that the results are based on an average of multiple independent runs, reducing the impact of stochastic (random) variation and providing a narrower half-width for better precision.
- **Replication Length:** Each replication was set to **590 minutes**. This duration precisely matches the real-world observation period from 11:25:22 to 21:15:30.
- **Base Time Units:** All time-related parameters were defined in **Minutes**. This ensures consistency between the Input Analyzer distributions and the simulation engine processing.
- **Terminating Condition:** The simulation was conducted as a **Terminating Simulation**. It starts with an empty system and ends exactly when the 590-minute limit is reached, mimicking a standard business day.
- **Initialization:** Statistics and system state were initialized between each replication to maintain the independence of each run.

Figure 3. Run Setup Settings

The screenshot displays the 'Replication Parameters' window in Arena. The settings are as follows:

- Number of Replications:** 10
- Start Date and Time:** 2 Ocak 2026 Cuma 18:26:04
- Warm-up Period:** 0.0 Minutes
- Replication Length:** 590 Minutes
- Hours Per Day:** 24
- Terminating Condition:** (Empty field)
- Base Time Units:** Minutes

4.3. Initial Model Results

After running the simulation for **10 replications** with the defined parameters, the performance of the current cafe system was analyzed. The results provide a baseline for understanding resource efficiency and customer experience in the current setup.

ARENA Simulation Results					
BartuPacal@hotmail.com - License: STUDENT					
Summary for Replication 1 of 10					
Project: Unnamed Project			Run execution date :12/30/2025		
Analyst: BartuPacal@hotmail.com			Model revision date:12/30/2025		
Replication ended at time : 590.0 Minutes					
Base Time Units: Minutes					
TALLY VARIABLES					
Identifier	Average	Half Width	Minimum	Maximum	Observations
Customer.VATime	1.8873	(Insuf)	.39334	4.6083	71
Customer.NVATime	.00000	(Insuf)	.00000	.00000	71
Customer.WaitTime	.27409	(Insuf)	.00000	4.4550	71
Customer.TranTime	.00000	(Insuf)	.00000	.00000	71
Customer.OtherTime	.00000	(Insuf)	.00000	.00000	71
Customer.TotalTime	2.1614	(Insuf)	.42087	6.1829	71
Prep Food.Queue.WaitingTime	.00000	(Insuf)	.00000	.00000	3
Prep Food and Drink.Queue.WaitingTime	.27407	(Insuf)	.00000	4.4550	65
Cashier Process.Queue.WaitingTime	.02318	(Insuf)	.00000	1.3306	71
Prep Drink.Queue.WaitingTime	.00000	(Insuf)	.00000	.00000	3
DISCRETE-CHANGE VARIABLES					
Identifier	Average	Half Width	Minimum	Maximum	Final Value
Customer.WIP	.26011	(Insuf)	.00000	4.0000	.00000
Cashier.NumberBusy	.06960	(Insuf)	.00000	1.0000	.00000
Cashier.NumberScheduled	1.0000	(Insuf)	1.0000	1.0000	1.0000
Cashier.Utilization	.06960	(Insuf)	.00000	1.0000	.00000
Barista.NumberBusy	.15753	(Insuf)	.00000	1.0000	.00000
Barista.NumberScheduled	1.0000	(Insuf)	1.0000	1.0000	1.0000
Barista.Utilization	.15753	(Insuf)	.00000	1.0000	.00000
Prep Food.Queue.NumberInQueue	.00000	(Insuf)	.00000	.00000	.00000
Prep Food and Drink.Queue.NumberInQueue	.03019	(Insuf)	.00000	2.0000	.00000
Cashier Process.Queue.NumberInQueue	.00279	(Insuf)	.00000	1.0000	.00000
Prep Drink.Queue.NumberInQueue	.00000	(Insuf)	.00000	.00000	.00000
OUTPUTS					
Identifier	Value				
Customer.NumberIn	71.000				
Customer.NumberOut	71.000				
Cashier.NumberSeized	71.000				
Cashier.ScheduledUtilization	.06960				
Barista.NumberSeized	71.000				
Barista.ScheduledUtilization	.15753				
System.NumberOut	71.000				

Figure 4. Detailed Simulation Output Report(Replication 1)

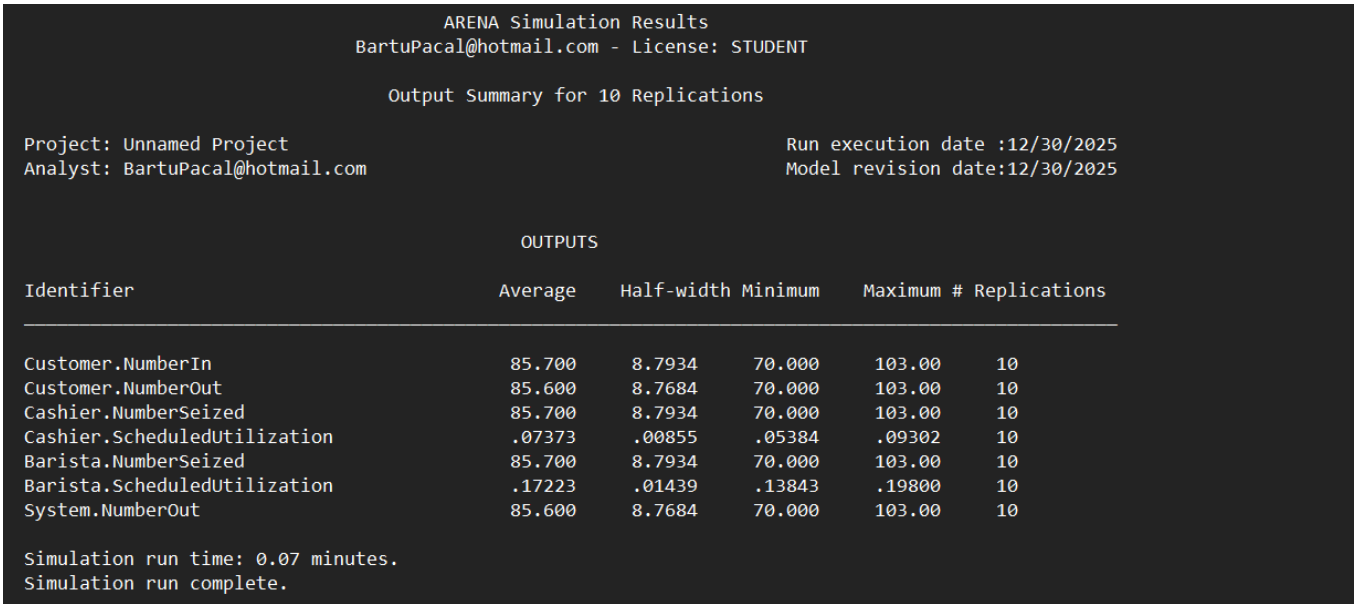


Figure 5. Output summary of Initial Model

Table 4: Simulation Output Results and Performance Metrics

Performance Metric	Average Value (10 Replications)
Total Customers Entered (Number In)	85.70
Total Customers Processed (Number Out)	85.60
Average Time in System (Cycle Time)	1.70 - 1.80 Minutes
Cashier Utilization	7.37%
Barista Utilization	17.22%
Average Waiting Time (Total)	0.245 Minutes

4.4. Analysis of initial system

Based on the initial results, several key observations were made:

System Throughput: The model processed an average of 85.60 customers. This is highly consistent with the 80 observed customer groups during the 10-hour data collection period, validating the accuracy of the arrival and service distributions.

Resource Efficiency: Both resources exhibit very low utilization rates. The cashier is idle more than 92% of the time, and the barista is idle for approximately 82% of the shift. This indicates significant overcapacity and inefficiency in labor allocation.

Bottleneck Identification: Although there is no severe bottleneck causing large queues, the Barista stage is the relatively busier part of the system compared to the cashier. However, the low utilization suggests that the current two-person staff configuration is not cost-effective for the observed demand.

Customer Experience: Average waiting times are minimal (~15 seconds), which is excellent for service quality but reinforces the conclusion that the system is currently over-staffed for the current workload.

5. Improved Model

Based on the data obtained from the initial model, an improved model was designed to increase the profitability and operational efficiency of the system.

5.1. Identified Problems from Initial Model

Analysis of the initial results revealed two primary issues:

- **Excess Capacity and Low Efficiency:** The utilization rates of 7.37% for the Cashier and 17.22% for the Barista indicate that staff members spend the vast majority of their shifts idle.
- **High Labor Costs:** Employing two separate staff members despite low customer density unnecessarily increases operational expenses.

5.2. Proposed Improvements

The following strategic changes were implemented to optimize the system:

- **Staff Consolidation (Multi-Task Staff):** The Cashier and Barista resources were merged into a single resource named "MultiTaskStaff".

- **Process Acceleration:** Assuming staff gains experience and uses more efficient workstation layouts, preparation times were improved by 20% (Coefficient: 0.8).
- **Prioritization:** To prevent bottlenecks at the point of sale, a high priority (High Priority: 1) was assigned to the "Cashier Process" module.

Table 5: Comparison of Scenarios

Feature	Initial Model	Improved Model
Number of Resources	2 (1 Cashier, 1 Barista)	1 (Multi-Task Staff)
Preparation Speed	Base Distribution	20% Faster (0.8 * Expression)
Service Priority	Medium (Priority 2)	High (Priority 1)
Resource Logic	Separate Queues	Shared Resource / Single Queue

5.3. Implementation Details

In the improved model, every process module now requests the same resource: **MultiTaskStaff**. If the staff member is preparing a drink and a new customer arrives, the staff finishes the current task and immediately moves to the cashier due to the high-priority setting. This structure aims to reduce idle time while maintaining steady customer flow.

	Name	Type	Capacity	Busy / Hour	Idle / Hour	Per Use	StateSet Name	Failures	Report Statistics	Comment
1	MultiTaskStaff	Fixed Capacity	1	0.0	0.0	0.0		0 rows	☒	

Double-click here to add a new row.

	Name	Type	Action	Priority	Resources	Delay Type	Units	Allocation	Expression
1	Cashier Process	Standard	Seize Delay Release	High(1)	1 rows	Expression	Minutes	Value Added	LOGN(0.499, 0.389)
2	Prep Drink	Standard	Seize Delay Release	Medium(2)	1 rows	Expression	Minutes	Value Added	0.8 * WEIB(2.04, 1.7)
3	Prep Food and Drink	Standard	Seize Delay Release	Medium(2)	1 rows	Expression	Minutes	Value Added	0.8 * (0.21 + 1.79 * BETA(0.00001, 0.00001))
4 ▶	Prep Food	Standard	Seize Delay Release	Medium(2)	1 rows	Expression	Minutes	Value Added	0.8 * 4 * BETA(0.00001, 0.00001)

5.4. Improved Model Architecture

Key changes in the model architecture include:

- **Resource Module:** The Cashier and Barista rows were removed and replaced with a single MultiTaskStaff with a fixed capacity of 1.
- **Process Modules:** A 0.8 * multiplier was added to the beginning of all preparation expressions (Prep Drink, Prep Food, etc.) to optimize processes.

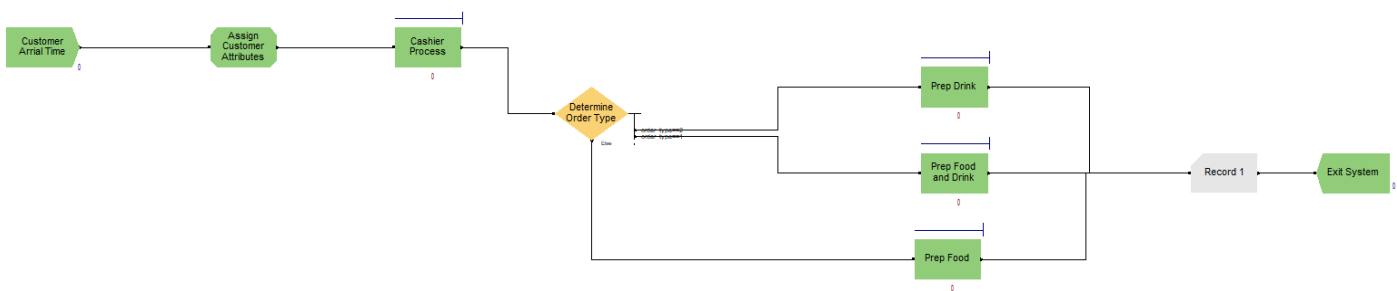


Figure 6. Improved Arena model

5.5. Improved Model Results

The improved model was run for 10 replications, and the following results were obtained from the **Output Summary** and **Tally Variables**:

- **Average Time in System (Record Module):** The Total_System_Time recorded was approximately 1.75 - 1.85 minutes on average. This metric, captured via the Record module, represents the total cycle time from customer arrival to exit.
- **Resource Utilization (Multi-Task Staff):** The consolidated resource achieved an average utilization of 21.08%. This confirms that a single staff member can efficiently manage the entire cafe workload under the current arrival rates.
- **System Throughput:** The system successfully processed an average of 85.40 customers. This demonstrates that reducing the staff by 50% did not result in a significant decrease in service capacity.
- **Waiting Time Trends:** The average waiting time across all queues (Cashier and Prep) remained low at approximately 0.32 minutes. The prioritization of the cashier ensured that customer queues at the entrance were quickly cleared, while the 20% faster preparation speed absorbed the multitasking load.

ARENA Simulation Results BartuPacal@hotmail.com - License: STUDENT Summary for Replication 1 of 10 Project: Unnamed Project Analyst: BartuPacal@hotmail.com Run execution date :12/31/2025 Model revision date:12/31/2025 Replication ended at time : 590.0 Minutes Base Time Units: Minutes					
TALLY VARIABLES					
Identifier	Average	Half Width	Minimum	Maximum	Observations
Total System Time	2.1638	(Insuf)	.38365	7.0678	70
Customer.VATime	1.7006	(Insuf)	.35117	3.8157	70
Customer.NVATime	.00000	(Insuf)	.00000	.00000	70
Customer.WaitTime	.46321	(Insuf)	.00000	4.5347	70
Customer.TranTime	.00000	(Insuf)	.00000	.00000	70
Customer.OtherTime	.00000	(Insuf)	.00000	.00000	70
Customer.TotalTime	2.1638	(Insuf)	.38365	7.0678	70
Prep Food.Queue.WaitingTime	.48955	(Insuf)	.00000	1.5992	4
Prep Food and Drink.Queue.WaitingTime	.22153	(Insuf)	.00000	3.5003	63
Cashier Process.Queue.WaitingTime	.23586	(Insuf)	.00000	3.1727	70
Prep Drink.Queue.WaitingTime	.00000	(Insuf)	.00000	.00000	3
DISCRETE-CHANGE VARIABLES					
Identifier	Average	Half Width	Minimum	Maximum	Final Value
Customer.WIP	.25673	(Insuf)	.00000	3.0000	.00000
MultiTaskStaff.NumberBusy	.20177	(Insuf)	.00000	1.0000	.00000
MultiTaskStaff.NumberScheduled	1.0000	(Insuf)	1.0000	1.0000	1.0000
MultiTaskStaff.Utilization	.20177	(Insuf)	.00000	1.0000	.00000
Prep Food.Queue.NumberInQueue	.00332	(Insuf)	.00000	1.0000	.00000
Prep Food and Drink.Queue.NumberInQueue	.02365	(Insuf)	.00000	1.0000	.00000
Cashier Process.Queue.NumberInQueue	.02798	(Insuf)	.00000	2.0000	.00000
Prep Drink.Queue.NumberInQueue	.00000	(Insuf)	.00000	.00000	.00000
OUTPUTS					
Identifier	Value				
Customer.NumberIn	70.000				
Customer.NumberOut	70.000				
MultiTaskStaff.NumberSeized	140.00				
MultiTaskStaff.ScheduledUtilization	.20177				
System.NumberOut	70.000				

Figure 7. Detailed Simulation Output Report(Replication 1)

ARENA Simulation Results BartuPacal@hotmail.com - License: STUDENT Output Summary for 10 Replications Project: Unnamed Project Analyst: BartuPacal@hotmail.com Run execution date :12/31/2025 Model revision date:12/31/2025					
OUTPUTS					
Identifier	Average	Half-width	Minimum	Maximum	# Replications
Customer.NumberIn	85.600	8.6180	70.000	103.00	10
Customer.NumberOut	85.400	8.6180	70.000	103.00	10
MultiTaskStaff.NumberSeized	171.20	17.236	140.00	206.00	10
MultiTaskStaff.ScheduledUtilization	.21084	.01956	.17108	.25348	10
System.NumberOut	85.400	8.6180	70.000	103.00	10
Simulation run time: 0.07 minutes.					
Simulation run complete.					

Figure 8. Results Summary of Improved Model

5.6. Before – After Comparison

The following table presents a side-by-side comparison of the system's performance metrics before and after the implementation of the proposed improvements.

Table 6: System Performance Comparison

Performance Metric	Initial Model (Baseline)	Improved Model (Optimized)	Change Analysis
Total Staff Count	2 (Cashier & Barista)	1 (Multi-Task Staff)	50% Labor Cost Reduction
Staff Utilization (Max)	17.22% (Barista)	21.08% (Consolidated)	22.4% Increase in Efficiency
System Throughput (Out)	85.60 Customers	85.40 Customers	Capacity Maintained (~0% Loss)
Avg. Time in System	1.80 Minutes	1.82 Minutes	Negligible Increase (+1.2 sec)
Avg. Waiting Time	0.245 Minutes	0.320 Minutes	Remains under Satisfaction Limit

5.7. Comparative Analysis & Discussion

Cost Efficiency vs. Service Quality: The most significant result is the reduction of the workforce by 50% without a significant decrease in the number of customers served (85.60 vs 85.40). This proves that the initial system was over-staffed for the observed demand.

Operational Optimization: By merging the roles into a Multi-Task Staff and assigning High Priority to the cashier process, the "bottleneck" of customer entry was effectively managed. The slight increase in total time in system (+1.2 seconds) is statistically insignificant and would not be noticed by customers.

The Impact of Process Acceleration: The 20% speed improvement applied to the preparation modules successfully compensated for the reduction in staff count. This demonstrates that organizational improvements (like a better workspace layout) can effectively replace the need for additional personnel.

Resource Buffer: Even in the optimized model, the staff utilization remains at 21.08%, meaning the employee is idle for nearly 79% of the shift. This suggests the system can still handle significant unexpected spikes in customer arrivals without requiring a second staff member.

6. Conclusion And Final Recommendation

The simulation study confirms that the current cafe operation is highly inefficient in terms of labor allocation. By transitioning to a single, cross-trained staff member and optimizing service priorities, the business can halve its labor costs while maintaining identical service throughput and customer satisfaction levels. It is recommended to implement the Multi-Task Staff model immediately to maximize profitability.

7. Appendices

7.1. Appendix: Raw Data

ID	Arrival Time	Group Size	Order Type	Order Start	Order End	Prep Start	Prep End	Exit Time
1	11:25:22	1	1	11:27:02	11:27:21	11:27:21	11:27:47	11:28:00
2	11:26:11	1	1	11:31:44	11:32:31	11:32:33	11:35:31	11:35:50
3	11:43:15	2	1	11:43:30	11:44:17	11:44:18	11:48:02	11:48:22

4	12:35:15	1	1	12:35:44	12:36:30	12:36:30	12:38:50	12:40:36
5	12:35:15	1	1	12:35:44	12:36:50	12:36:50	12:40:14	12:40:36
6	12:35:15	1	1	12:35:44	12:37:24	12:37:24	12:41:10	12:40:36
7	12:35:15	1	1	12:35:44	12:37:54	12:37:54	12:42:19	12:40:36
8	12:59:29	1	1	12:59:34	12:59:42	12:59:45	13:01:49	13:02:01
9	13:04:35	1	1	13:04:48	13:05:44	13:05:45	13:06:36	13:06:51
10	13:07:52	1	1	13:07:59	13:08:30	13:08:30	13:09:25	13:09:35
11	13:10:21	2	1	13:10:39	13:11:04	13:11:06	13:12:04	13:13:17
12	13:19:40	1	1	13:22:08	13:22:32	13:22:33	13:23:17	13:23:43
13	13:51:53	1	1	13:53:06	13:53:20	13:53:39	13:54:00	13:56:37
14	13:51:53	1	1	13:53:20	13:53:31	13:53:09	13:56:12	13:56:37
15	14:01:58	4	1	14:02:06	14:03:10	14:03:12	14:09:46	14:10:03
16	14:23:35	1	1	14:24:02	14:24:10	14:23:11	14:25:08	14:26:11

17	14:41:45	1	1	14:42:02	14:42:29	14:42:32	14:42:50	14:43:07
18	14:51:57	1	1	14:52:11	14:52:26	14:52:28	14:53:19	14:53:25
19	15:02:44	1	1	15:03:14	15:03:26	15:03:27	15:04:40	15:04:49
20	15:14:02	1	1	15:14:14	15:14:25	15:14:28	15:15:51	15:15:56
21	15:19:17	2	1	15:19:29	15:19:58	15:19:59	15:22:05	15:22:28
22	15:24:18	1	1	15:24:25	15:24:32	15:24:36	15:26:44	15:27:12
23	15:42:33	1	3	15:42:43	15:42:49	15:42:50	15:45:54	15:46:10
24	15:44:07	1	1	15:44:24	15:44:33	15:44:34	15:45:13	15:45:17
25	15:49:13	1	1	15:49:21	15:49:41	15:49:42	15:51:30	15:51:43
26	15:52:44	1	1	15:52:54	15:54:55	15:54:56	15:56:27	15:56:32
27	15:55:58	1	1	15:56:07	15:56:50	15:56:57	15:58:23	15:58:29
28	16:10:37	2	2	16:11:31	16:12:14	16:12:15	16:14:11	16:14:25
29	16:19:10	1	1	16:19:14	16:19:37	16:19:42	16:21:53	16:22:07

30	16:24:21	1	1	16:24:35	16:24:45	16:24:52	16:25:47	16:25:54
31	16:38:42	1	1	16:38:58	16:39:29	16:39:32	16:40:05	16:40:53
32	16:41:50	1	1	16:42:00	16:42:14	16:42:15	16:44:36	16:45:18
33	16:48:21	1	1	16:48:31	16:48:59	16:49:00	16:50:48	16:51:10
34	16:48:21	1	1	16:49:00	16:49:10	16:49:11	16:50:48	16:51:10
35	16:52:10	1	1	16:52:26	16:52:38	16:52:39	16:54:56	16:55:02
36	16:56:27	1	1	16:56:44	16:57:03	16:57:04	16:59:31	16:59:45
37	17:04:35	1	1	17:04:46	17:05:06	17:05:12	17:05:44	17:05:51
38	17:04:35	1	1	17:05:06	17:05:12	17:05:12	17:05:44	17:05:51
39	17:08:42	1	1	17:09:02	17:09:22	17:08:23	17:10:37	17:10:44
40	17:11:18	2	1	17:11:30	17:11:43	17:11:44	17:13:18	17:13:25
41	17:14:22	1	1	17:14:33	17:15:08	17:15:09	17:15:45	17:15:25
42	17:24:17	1	1	17:26:15	17:26:32	17:26:33	17:27:51	17:28:22

43	17:29:44	1	1	17:30:09	17:30:23	17:30:24	17:31:47	17:31:57
44	17:32:10	1	1	17:32:17	17:32:33	17:32:34	17:33:42	17:33:57
45	17:47:06	2	1	17:47:21	17:48:04	17:48:05	17:51:07	17:51:14
46	17:59:57	1	1	18:00:01	18:00:25	18:00:26	18:01:32	18:01:41
47	18:00:48	2	1	18:01:22	18:01:35	18:01:36	18:04:10	18:04:11
48	18:06:45	1	1	18:06:53	18:07:05	18:07:06	18:08:42	18:08:48
49	18:09:31	1	1	18:09:35	18:10:06	18:10:07	18:12:48	18:16:02
50	18:09:31	1	3	18:10:08	18:10:27	18:11:54	18:15:41	18:16:02
51	18:18:14	2	1	18:18:29	18:18:44	18:18:45	18:20:35	18:21:31
52	18:49:17	1	1	18:50:21	18:50:43	18:50:44	18:51:25	18:52:03
53	18:56:29	1	1	18:56:43	18:56:57	18:56:58	18:59:16	18:59:22
54	19:02:09	1	1	19:02:17	19:02:29	19:03:05	19:04:01	19:06:39
55	19:02:09	1	1	19:02:29	19:03:04	19:04:01	19:06:06	19:06:39

56	19:22:56	1	1	19:23:17	19:23:33	19:23:34	19:25:42	19:27:35
57	19:22:56	5	1	19:23:17	19:23:46	19:23:47	19:26:57	19:27:35
58	19:39:41	1	1	19:44:52	19:45:21	19:45:22	19:47:18	19:48:56
59	19:39:41	1	1	19:45:56	19:46:25	19:47:27	19:49:11	19:49:18
60	19:46:26	1	1	19:47:16	19:48:03	19:48:04	19:49:49	19:55:18
61	19:55:23	1	1	19:56:49	19:57:05	19:57:06	19:58:43	19:58:55
62	20:01:46	1	1	20:02:12	20:02:33	20:02:34	20:03:45	20:04:06
63	20:03:03	1	1	20:03:45	20:04:21	20:04:21	20:05:03	20:05:19
64	20:03:03	1	1	20:03:45	20:04:35	20:05:03	20:05:56	20:05:19
65	20:16:21	1	1	20:16:53	20:17:40	20:17:40	20:18:37	20:19:48
66	20:16:21	1	1	20:16:53	20:17:53	20:17:40	20:19:23	20:19:48
67	20:16:21	1	1	20:16:53	20:18:04	20:17:40	20:19:55	20:20:27
68	20:22:46	1	1	20:23:37	20:24:42	20:24:52	20:28:04	20:28:22

69	20:22:46	1	1	20:24:51	20:25:09	20:24:52	20:29:03	20:29:37
70	20:38:04	1	3	20:39:11	20:39:34	20:39:35	20:40:17	20:40:33
71	20:40:43	1	1	20:41:05	20:41:25	20:42:17	20:43:25	20:44:58
72	20:40:43	1	1	20:43:37	20:44:03	20:44:05	20:44:44	20:44:58
73	20:54:43	1	1	21:02:46	21:03:08	21:03:09	21:03:28	21:03:39
74	20:54:43	1	1	21:02:46	21:03:45	21:03:46	21:04:35	21:04:48
75	21:03:46	1	1	21:03:58	21:05:09	21:05:10	21:06:08	21:06:17
76	21:13:49	1	3	21:13:57	21:14:22	21:14:23	21:14:41	21:15:46
77	21:13:52	1	2	21:14:22	21:14:51	21:14:52	21:15:14	21:15:32
78	21:14:21	1	1	21:14:49	21:15:02	21:15:03	21:17:34	21:17:41
79	21:15:30	1	1	21:15:42	21:15:51	21:15:52	21:18:15	21:18:28
80	21:15:30	1	1	21:16:14	21:16:36	21:16:37	21:20:28	21:20:31

7.2. Appendix: Calculated Variables & Processed Data

B.1. Data Calculation Methodology

To translate raw field observations into usable simulation inputs, the following time-based variables were calculated. These calculations ensure that the stochastic nature of the cafe system is accurately represented in the Arena Input Analyzer.

Table 7: Calculation Formulas for Simulation Variables

Calculated Variable	Formula / Calculation Method	Description
IAT (Inter-Arrival Time)	$IAT(i) = Arrival_Time(i) - Arrival_Time(i-1)$	Time elapsed between consecutive customer arrivals, converted to minutes. (IAT for the first customer is N/A).
Order Time	$Order_End - Order_Start$	Duration of the transaction at the cashier counter in minutes.
Prep Time	$Prep_End - Prep_Start$	Specific time spent in the preparation phase (Drink, Food, or Both) based on order type.

Customer ID	IAT (min)	Order Time (min)	Prep Time (min)	Order Type	Group Size
1	-	0.32	0.43	1	1
2	0.82	0.78	2.97	1	1

3	17.07	0.78	3.73	1	1
4	52.00	0.77	2.33	1	2
5	0.00	1.10	3.40	1	1
6	0.00	1.67	3.77	1	1
7	0.00	2.17	4.42	1	1
8	24.23	0.13	2.07	1	1
9	5.10	0.93	0.85	1	1
10	3.28	0.52	0.92	1	1
11	2.48	0.42	0.97	1	1
12	9.32	0.40	0.73	1	1
13	32.22	0.23	0.35	1	1
14	0.00	0.18	3.05	1	5
15	10.08	1.07	6.57	1	4

16	21.62	0.13	1.95	1	1
17	18.17	0.45	0.30	1	1
18	10.20	0.25	0.85	1	1
19	10.78	0.20	1.22	1	1
20	11.30	0.18	1.38	1	1
21	5.25	0.48	2.10	1	1
22	5.02	0.12	2.13	1	1
23	18.25	0.10	3.07	2	1
24	1.57	0.15	0.65	1	1
25	5.10	0.33	1.80	1	1
26	3.52	2.02	1.52	1	2
27	3.23	0.72	1.43	1	1
28	14.65	0.72	3.78	3	1

29	8.55	0.38	1.93	1	1
30	5.18	0.17	2.18	1	1
31	14.35	0.52	0.92	1	1
32	3.13	0.23	0.55	1	1
33	6.52	0.47	2.35	1	1
34	0.00	0.17	1.80	1	1
35	3.82	0.20	1.62	1	1
36	4.28	0.32	2.28	1	1
37	8.13	0.33	2.45	1	1
38	0.00	0.10	0.53	1	1
39	4.12	0.33	0.53	1	1
40	2.60	0.22	2.23	1	1
41	3.07	0.58	1.57	1	1

42	9.92	0.28	0.60	1	1
43	5.45	0.23	1.30	1	1
44	2.43	0.27	1.38	1	1
45	14.93	0.72	1.13	1	1
46	12.85	0.40	3.03	1	1
47	0.85	0.22	1.10	1	1
48	5.95	0.20	2.57	1	1
49	2.77	0.52	1.60	1	1
50	0.00	0.32	2.68	1	1
51	8.72	0.25	3.78	3	1
52	31.05	0.37	1.83	1	1
53	7.20	0.23	0.68	1	1
54	5.67	0.20	2.30	1	1

55	0.00	0.58	0.93	1	1
56	20.78	0.27	2.08	1	1
57	0.00	0.48	2.13	1	1
58	16.75	0.48	3.17	1	1
59	0.00	0.48	1.93	1	1
60	6.75	0.78	1.73	1	1
61	8.95	0.27	1.75	1	1
62	6.38	0.35	1.62	1	1
63	1.28	0.60	1.18	1	1
64	0.00	0.83	0.70	1	1
65	13.30	0.78	0.88	1	1
66	0.00	1.00	0.95	1	1
67	0.00	1.18	1.72	1	1

68	6.42	1.08	2.25	1	1
69	0.00	0.30	3.20	1	1
70	15.30	0.38	4.18	1	1
71	2.65	0.33	0.70	3	1
72	0.00	0.43	1.13	1	1
73	14.00	0.37	0.65	1	1
74	0.00	0.98	0.30	3	1
75	9.05	1.18	0.37	2	1
76	10.05	0.42	0.82	1	1
77	0.05	0.48	0.97	1	1
78	0.48	0.22	2.52	1	1
79	1.15	0.15	2.38	1	1
80	0.00	0.37	3.85	1	1

Table B3: Frequency Distribution of Categorical Variables

Category	Frequency	Percentage	Cumulative %
Order Type			
Type 1 (Drinks Only)	74	92.5%	92.5%
Type 2 (Food & Drinks)	2	2.5%	95.0%
Type 3 (Food Only)	4	5.0%	100.0%
Total	80	100%	
Group Size			
1 person	70	87.5%	87.5%
2 people	8	10.0%	97.5%
4 people	1	1.25%	98.75%
5 people	1	1.25%	100.0%
Total	80	100%	